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1. Compute the following limits in DETAIL and WITHOUT USING DERIVATIVE(!)

$$\frac{1}{\infty} = 0 \quad \frac{1}{x} = \alpha$$

$$\bullet \lim_{x \rightarrow \infty} \frac{\sin\left(\frac{1}{x}\right)}{\frac{1}{x}} = \lim_{\alpha \rightarrow 0} \frac{\sin \alpha}{\alpha} = 1$$

$$\frac{\sin 0}{0} = \frac{0}{0}$$

$$\bullet \lim_{(x \rightarrow 1^+)} \frac{|x-1| + \lfloor x-3 \rfloor}{x+1} = \frac{x-1 + \lfloor x-3 \rfloor}{x+1} = \frac{0-2}{2} = -1$$

$$x=1^+ \Rightarrow |x-1| > 0$$

$$x \rightarrow 1^+ \quad \lfloor x-3 \rfloor \Rightarrow -2$$

$$\bullet \lim_{x \rightarrow -\infty} \frac{\sqrt{5x^2+3x-2}}{3x+5} = \frac{\sqrt{x^2(5+\frac{3}{x}-\frac{2}{x^2})}}{x(3+\frac{5}{x})} = \frac{|x|\sqrt{5}}{x \cdot 3} = \frac{-x\sqrt{5}}{x \cdot 3} = -\frac{\sqrt{5}}{3}$$

$$\lim_{x \rightarrow -\infty} |x| < 0$$

$$\odot \lim_{x \rightarrow 0^+} \frac{1-\cos x}{x} = \frac{1-\cos x}{x} \cdot \frac{1+\cos x}{1+\cos x} = \frac{1-\cos^2 x}{x(1+\cos x)} = \frac{\sin^2 x}{x(1+\cos x)} = \frac{\sin x}{1+\cos x} \cdot \frac{\sin x}{x}$$

$$\frac{1-1}{0} = \frac{0}{0} \quad \Rightarrow \quad \frac{\sin x}{2x} \cdot \sin x = \frac{1}{2} \cdot 0 = \frac{0}{2}$$

$$\bullet \lim_{x \rightarrow -\infty} \frac{3x-\sqrt{5}}{\sqrt{9x^2+3x+1}} = \frac{x(3-\frac{\sqrt{5}}{x})}{|x|\sqrt{9+\frac{3}{x}+\frac{1}{x^2}}} = \frac{x \cdot 3}{-x \cdot 3} = -1$$

$$\frac{1}{\infty} = 0$$

$$\lim_{x \rightarrow -\infty} |x| < 0$$

$$\bullet \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2+x+1}}{2x+5} = \frac{\sqrt{x^2(1+\frac{1}{x}+\frac{1}{x^2})}}{x(2+\frac{5}{x})} = \frac{|x|}{2x} = \frac{-x}{2x} = -\frac{1}{2}$$

$$\lim_{x \rightarrow -\infty} |x| < 0$$

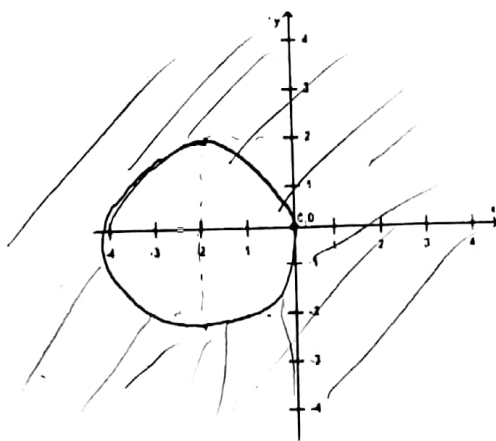
$$\bullet \lim_{(x \rightarrow 1^-)} \frac{|x-1| + \lfloor x \rfloor + 2x}{\sqrt{3}x} = \frac{(1-x) + \lfloor x \rfloor + 2x}{\sqrt{3}x} = \frac{0+0+2}{\sqrt{3}} = \frac{2}{\sqrt{3}}$$

$$x \rightarrow 1^- \Rightarrow |x-1| < 0$$

$$x \rightarrow 1^- \quad \lfloor x \rfloor = 0$$

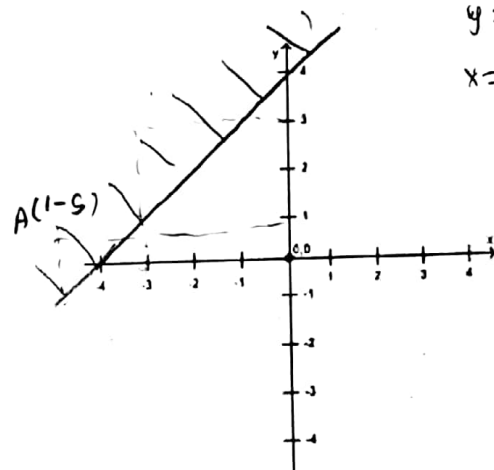
2. Sketch the graphs of the following relations on right coordinate system.

(a) $R = \{(x, y) | \sqrt{(x+2)^2 + y^2} > \sqrt{4}\}$



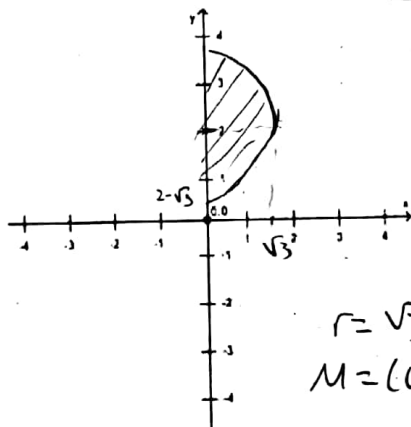
$r=2$
 $M=(-2, 0)$

(b) $S = \{(x, y) | y > x + 4\}$



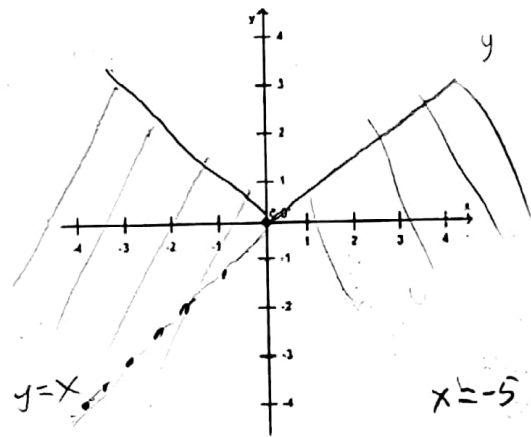
$y=0 \Rightarrow x=-4$
 $x=0 \Rightarrow y=4$

(c) $T = \{(x, y) | \sqrt{x^2 + (y-2)^2} \leq \sqrt{3}, x \geq 0\}$



$r=\sqrt{3}$
 $M=(0, 2)$

(d) $U = \{(x, y) | y \leq |x|\}$



$|x| < 0$ $y \leq -x$ $x=-5 \Rightarrow y=5$
 $|x| > 0$ $y \leq x$ $x=5 \Rightarrow y=5$

3. Find the solution set of the following inequality analyzing it in two cases ($2x + 3 \neq 0$).

$\frac{3x-2}{2x+3} > 0$

Case I: $2x + 3 < 0$

$2x < -3 \Rightarrow x < -\frac{3}{2}$

$x-5 < 0$

$x < 5$



$(-\infty, -\frac{3}{2})$

$\frac{x-5}{2x+3} > 0$

$\frac{3x-2}{2x+3} > 1$

Case II: $2x + 3 > 0$

$2x > -3$

$x > -\frac{3}{2}$

$x-5 > 0$

$x > 5$



$(5, \infty)$